COMPASS → PEER — author×year career master table (PD29 / Alex lock)

Last synced: 2026-09-02

Audience: Charles → PEER (Rivanna). **Context:** Paper Directions 29 — Alex locked cumulative career pubs rate for tenure HERO; Mac needs a precomputed spine to rsync.

Related: `transcripts/PD29_notes.md` · `TENURE_hero_pipeline.md` § PD29 delta

One sentence

Build a **flat author×calendar-year table** (pubs + running cumulative + career-length rate) from existing OpenAlex works, so Mac (or Rivanna) can slice decision-year tenure analysis.

Mac-first (2026-09-02): `python tenure/tenure_pipeline/build_author_year_career_master.py` — ~2s, no snapshot. Rivanna Slurm optional for re-run after OA cache refresh.

Why (30 seconds)

Alex (Sep 2): tenure decision ≈ **year 6 up-or-out**; own performance = **cumulative pubs ÷ (focus_year − first_pub_year)**; career start = **first publication year** (OpenAlex). Mac should not recompute this from scratch every HERO run — one master table, then filter.

Inputs already on Rivanna (per May 2026 handoff):

File	Role
`tenure/tenure_pipeline/openalex_author_ids.jsonl`	Author universe + `faculty_id` + match tier
`tenure/tenure_pipeline/openalex_works_by_year.jsonl`	`{faculty_id, openalex_id, uni_slug, year, n_works}` per line

Do not re-run full HTML scrape or `build_openalex_cache.slurm` unless works file is missing/stale.

Deliverable

Primary output (pick one; prefer JSONL for consistency):
`tenure/tenure_pipeline/author_year_career_master.jsonl`

One row per `(openalex_id, year)` where `n_works > 0` **OR** year is on the author's career span through `max_year` in works (see schema).

Optional companion: `author_year_career_master_meta.json` (row counts, build timestamp, source file mtimes).

Schema (required columns)

Column	Type	Definition
`openalex_id`	str	OpenAlex author ID
`faculty_id`	str	From author_ids join (empty if unmatched)
`match_confidence`	str	HIGH / MEDIUM / … from author_ids
`year`	int	Calendar year
`n_works`	int	Publications that year (= `n_works` in works file; 0 allowed on padded years if you emit full span)
`cum_works`	int	Running sum of `n_works` through `year` (inclusive)
`first_pub_year`	int	min(year) with n_works > 0 for this author
`career_age_years`	int	`year - first_pub_year` (0 in first pub year)
`pubs_per_career_year`	float	`cum_works / career_age_years` when `career_age_years > 0`, else null

Alex denominator check: first pub year 0, decision year 10, 100 cum pubs → rate **10.0** (= 100 / (10−0), not ÷11).

Universe

- **All authors** appearing in `openalex_author_ids.jsonl` with `match_confidence` ∈ **{HIGH, MEDIUM}** (configurable env `CONFIDENCE_MIN`, default HIGH).
- Join works from `openalex_works_by_year.jsonl`; authors with ID but no works → omit or emit metadata-only line in meta JSON (PEER pick — document in meta).

Suggested implementation

1. **New script** (Mac can push via git before run):
   `tenure/tenure_pipeline/build_author_year_career_master.py`
   - Read-only on existing JSONL
   - Incremental write + flush per author (see project incremental-write rule)
   - Resume: skip authors already in output if re-run

2. **New Slurm wrapper** (mirror `build_openalex_cache.slurm`): `build_author_year_career_master.slurm` at repo root
   - `--mem=16G`, `--time=02:00:00`, `tenure_net` env
   - Lightweight vs OA snapshot scan
3. **Submit:**

```
cd ~/Ivy_Net && git pull origin main
sbatch build_author_year_career_master.slurm
tail -f slurm_out/slurm-build_author_year_career_master-*.out
```

4. **Charles pulls:**

```
./scripts/rsync_pull_recent_hpc.sh tenure
```

Preflight (PEER)

- [ ] `openalex_works_by_year.jsonl` exists and mtime ≥ last panel build
- [ ] Line count sanity: `wc -l openalex_works_by_year.jsonl openalex_author_ids.jsonl`
- [ ] If works stale vs new author IDs → run `sbatch build_openalex_cache.slurm` **first**, then career master

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Success criteria

1. `author_year_career_master.jsonl` on Rivanna under `tenure/tenure_pipeline/`
2. Meta documents: N authors, N author-year rows, year range, tier filter
3. Spot check: pick one known faculty_id — `first_pub_year`, `cum_works` monotonic, rate at last year matches hand calc
4. Charles rsync pull succeeds; file visible on Mac for COMPASS slice (decision-year cohort = **next** step, not this job)

Out of scope (this job)

- Decision-year flags / tenure outcome (Mac + `faculty_panel_with_pools.jsonl`)
- Department pond LOO at decision year
- Re-scrape / URL worksheet / Cell 2–4

Thread log

Date	Entry
2026-09-02	COMPASS draft after PD29; Charles to send PEER; build script TBD on git push.